

Multi-GlaucNet: A multi-task model for optic disc segmentation, blood vessel segmentation and glaucoma detection

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ABSTRACT

Glaucoma is a common and severe ocular disease that often leads to vision loss. The information on the optic disc (OD) and blood vessels in fundus images can significantly aid in glaucoma detection. In addition, the use of deep learning models for glaucoma detection is a highly effective approach. We propose a multi-task deep learning model called Multi-GlaucNet that can simultaneously segment the OD and blood vessels, thereby assisting doctors in diagnosing glaucoma. Multi-GlaucNet consists of three modules: the OD segmentation module, blood vessel segmentation module and glaucoma detection module. The OD segmentation module and blood vessel segmentation module both adopt an encoder-decoder structure to segment OD and blood vessels, respectively. The segmentation module is constructed using bottleneck layers in the encoding process and uses Pixel Shuffle and channel attention mechanisms in the decoding process. The detection module uses the ResNet50 network to perform glaucoma detection based on the features extracted from the segmentation modules. Multi-GlaucNet demonstrates outstanding performance in three areas. It achieves a Dice coefficient of 96.7% for OD segmentation on the ORIGA dataset, accuracy of 0.9798 and F1 score of 0.8562 for blood vessel segmentation on a mixed dataset. For glaucoma detection on the REFUGE dataset, it attains the highest accuracy of 0.967 and an area under the curve (AUC) of 0.950. These results validate the effectiveness of Multi-GlaucNet for glaucoma detection. The model's ability to perform multiple tasks with high accuracy and efficiency demonstrates that it is a valuable aiding tool for glaucoma diagnosis.

1. Introduction

Glaucoma is one of the leading causes of blindness in individuals aged 50 years and above. The global prevalence of glaucoma in this age group is estimated to be 3.6 million cases. The symptoms of glaucoma include blurred central vision, reduced peripheral vision, dark shadows in the field of vision, eye pain, headaches, nausea and vomiting [1]. In the early stage, glaucoma may not be present with noticeable symptoms. By the time patients experience symptoms, glaucoma has typically progressed, leading to reduced peripheral vision and vision loss [2]. Consequently, identifying glaucoma at an early stage is imperative to lessen its detrimental effects and avert harmful outcomes for patients.

Fundus images, which are captured by fundus imaging, can provide abundant information on eyes condition [3]. In these images, various components of the retina can be visualised clearly, allowing the rapid prediction of ocular diseases. The cup-to-disc ratio (CDR) is a commonly used method for identifying glaucoma, which represents the ratio

between the optic cup (OC) and OD [4,5]. However, different doctors may have varying levels of proficiency in judging glaucoma. In remote areas, there may also be a lack of sufficient medical resources for the detection of such diseases.

Crafted with precision for image data manipulation, convolutional neural networks (CNNs) are a specialised segment of deep learning models. CNNs have found extensive utilisation within the realm of computer-aided diagnosis (CAD), resulting in excellent enhancement in the precision and efficiency of CAD systems. Consequently, CNNs have emerged as indispensable tools for tasks such as OD segmentation, blood vessel delineation and even glaucoma detection, with numerous well-established applications substantiating their effectiveness in these domains [6–8].

Various deep learning models, each focusing on a single task, such as OD segmentation, blood vessel analysis or glaucoma detection, currently exist in the field. However, existing research on glaucoma is insufficiently explored in the context of multi-task deep learning models.

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Current approaches in this domain predominantly use pixel information from OD and OC segmentation, neglecting the valuable data encapsulated within blood vessels. A model capable of concurrently executing OD segmentation, blood vessel segmentation and glaucoma monitoring could provide crucial diagnostic information for CAD [9].

In this paper, we present Multi-GlaucNet, which is a novel multi-task deep learning model that can simultaneously perform OD and blood vessel segmentation and glaucoma detection. Multi-GlaucNet uses two independent segmentation modules for blood vessel and OD segmentation, generating segmentation masks as outputs. The segmentation module is an encoder–decoder architecture network model, where the encoder part is composed of bottleneck blocks. During the decoding process, Pixel Shuffle is used instead of traditional transposed convolution and a channel attention mechanism is used to enhance the decoding performance. The concluding step involves submitting both the original and masked images to the detection network, which then proceeds to detect glaucoma.

As a result, Multi-GlaucNet provides segmentation results for the OD and blood vessels, with an additional outcome of glaucoma detection. Using Multi-GlaucNet, one can obtain simultaneous and accurate segmentation of the OD and blood vessels, together with predictions for glaucoma. This multi-faceted analysis of fundus images contributes to an overall improvement in the diagnostic accuracy of glaucoma, ultimately providing a comprehensive approach to the disease diagnosis.

In this study, we compared Multi-GlaucNet with dedicated deep learning models for OD segmentation, blood vessel segmentation and glaucoma detection separately. Results demonstrate that Multi-GlaucNet outperforms these specialised models in all three tasks.

The remaining sections of this paper are organised as follows. In Section 2, we review relevant research on different tasks. Section 3 introduces the proposed Multi-GlaucNet framework. In Section 4, the experimental setup and results are presented. Finally, Section 5 includes our conclusions.

2. Related works

In this section, we will provide a brief review of the relevant literature related to the tasks involved in Multi-GlaucNet, including the following: the OD segmentation task, vessel segmentation task and glaucoma detection task.

2.1. OD segmentation task

OD segmentation can assist medical professionals in the automated detection and quantitative analysis of various diseases, such as glaucoma and diabetic retinopathy.

Shyamalee et al. [10] proposed an attention-enhanced U-Net model that incorporates attention gates at each skip connection to dynamically and selectively enhance useful features.

Meng Y et al. [11] also proposed a graph convolutional network-based method for the weakly supervised and semi-supervised segmentation of OD and OC, along with the estimation of the vertical CDR (vCDR), to assist in glaucoma assessment from fundus images.

Lu Shuai et al. [3] proposed a method called prior knowledge-based relationship transformer network (PKRT-Net) to assist OC and OD segmentation. PKRT-Net leverages clinical prior knowledge to aid in OC segmentation and effectively models long-range relationships of spatial features using a transformer model.

2.2. Glaucoma detection task

Glaucoma detection typically involves observing changes in OC, OD and blood vessels in fundus images. It provides vital visual information to medical professionals, assisting them in making accurate diagnoses and treatment decisions to safeguard patients' vision and ocular health.

Nawaz M et al. [12] proposed a deep learning-based method for

automated glaucoma detection using the localisation of the OD and OC. Their approach combines EfficientNet-B0 as the feature extractor and uses the bidirectional feature pyramid network as the feature fusion module.

Shyamalee et al. [13] used three distinct CNN architectures—Inception-v3, Visual Geometry Group 19 (VGG19) and Residual Neural Network 50 (ResNet50)—to classify fundus images for identifying glaucoma patients. In addition, various data pre-processing and augmentation techniques were used in the study to prevent overfitting and achieve high accuracy.

2.3. Multi-task learning

In the context of multi-task learning, comprehensive feature extraction from colour fundus images is achieved through various tasks. For disease monitoring and segmentation tasks, the distinct output characteristics of the two assignments pose additional challenges.

Hervella et al. [14] introduced a novel multi-task learning approach that concurrently performs segmentation of the OD and OC and classification of glaucoma in retinal images. By leveraging complementary information from both image-level and pixel-level annotations, the method enhances overall performance.

Pascal et al. [9] proposed a new method that uses related retinal imaging tasks and measurements to assist in the diagnosis of glaucoma. This approach uses a deep learning model to predict the presence of glaucoma in retinal images while simultaneously segmenting the OD and OC and localising the macula.

Shyamalee et al. [15] proposed a computational model for the segmentation and classification of retinal fundus images to detect glaucoma. To prevent overfitting, various data augmentation techniques were applied and several data pre-processing methods were used to enhance image quality and achieve high accuracy.

3. Methods

In this paper, we propose a multi-task deep learning model called Multi-GlaucNet, which is capable of simultaneously segmenting OD and blood vessels in fundus images and detecting glaucoma.

Fig. 1 illustrates the overall workflow of Multi-GlaucNet. Multi-GlaucNet is divided into three parts: OD segmentation, blood vessel segmentation and glaucoma classification.

In this end-to-end deep learning model, for a given fundus image, we first use an encoder–decoder segmentation module based on U-Net [16] to separately perform OD and vessel segmentation. The segment modules generate an OD segmentation mask and a vessel mask, which are then applied to the original image. The masked OD and vessel images, along with the original image, are then used as input to a ResNet50 [17] model for a binary detection task.

During the training phase, at first, we train the segmentation modules for OD and vessels separately. Once the training for segmentation is complete, we freeze the parameters of the two tasks to train the detection module.

3.1. Segmentation of OD and blood vessels

In the segmentation task, we used two separate segmentation networks to perform the segmentation of OD and blood vessels independently.

Before inputting the images into the model, we performed pre-processing on the fundus images. First, the images are resized to a resolution of 512×512 . Once images are cropped, we use contrast-limited adaptive histogram equalisation (CLAHE) for bolstering the contrast, which aids in making finer details more visible. CLAHE enhances image contrast by dividing the image into small blocks and performing histogram equalisation on each block separately. Fig. 2 shows fundus images before and after applying CLAHE. CLAHE is widely used in fundus

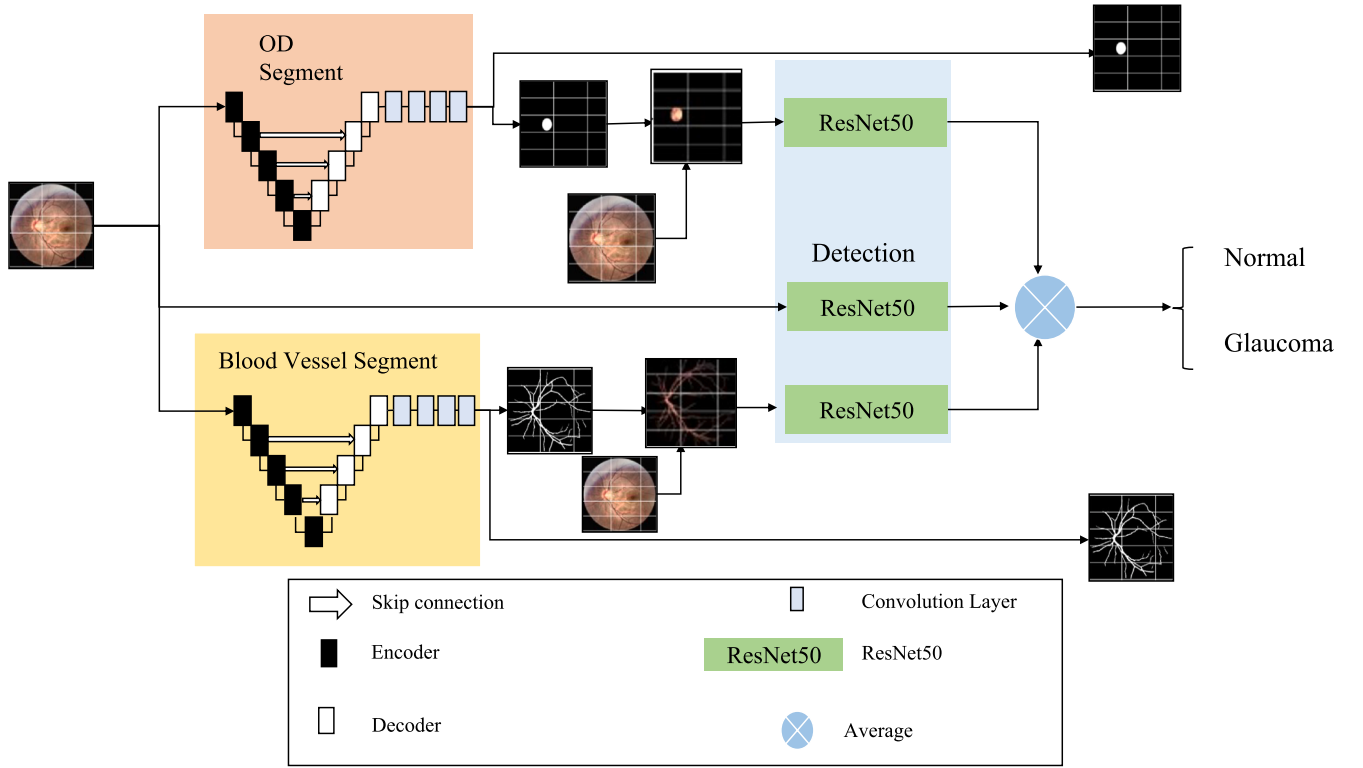


Fig. 1. Multi-GlaucNet architecture. Multi-GlaucNet is divided into three modules: OD segmentation, blood vessel segmentation and glaucoma detection. For the input fundus image, the two segmentation modules first segment the OD and blood vessels. Then, the input image, along with the segmented OD and blood vessel images, is fed into ResNet50 detection branches for glaucoma detection.

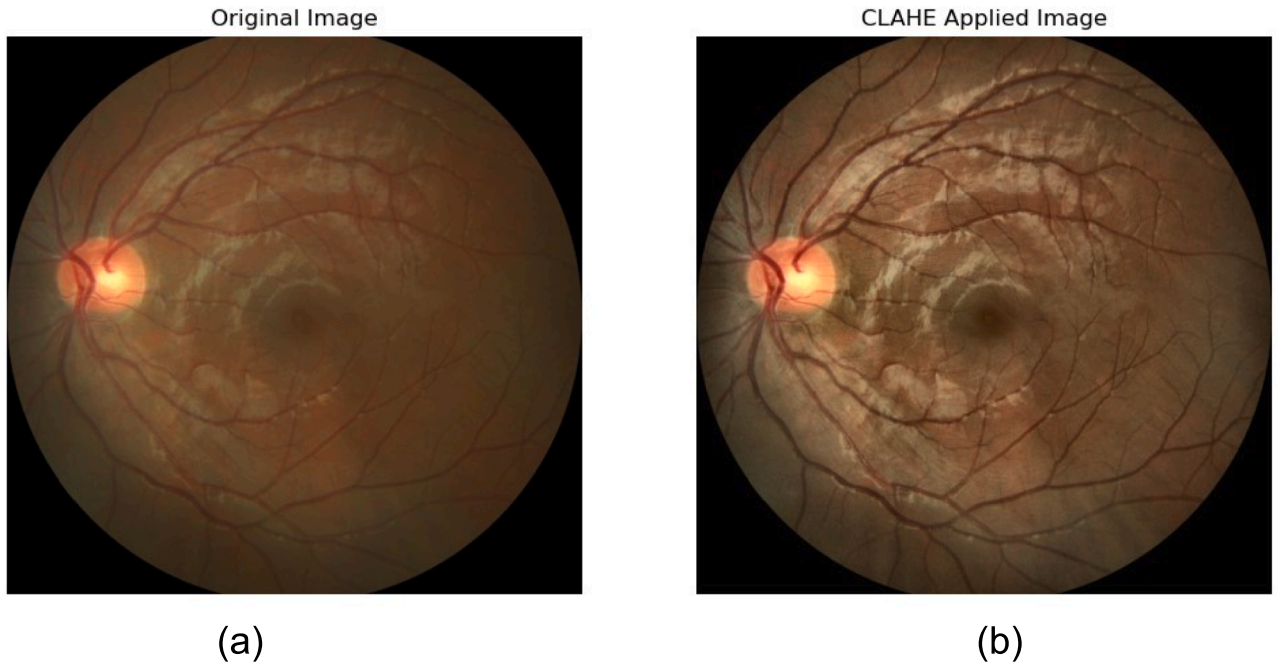


Fig. 2. Fundus images before and after CLAHE processing. (a) shows image before CLAHE processing and (b) shows image after CLAHE processing.

images to enhance the details of the blood vessels, macula and OD.

Fig. 3 illustrates the encoder-decoder structure used in the segmentation module of Multi-GlaucNet. The segmentation module is composed of a multi-layer encoder consisting of bottleneck blocks, a

multi-layer decoder composed of Pixel Shuffle and convolutional layers and an output convolutional layer. The input to the module is the augmented image, and the output is the corresponding greyscale mask of the OD or vessels.

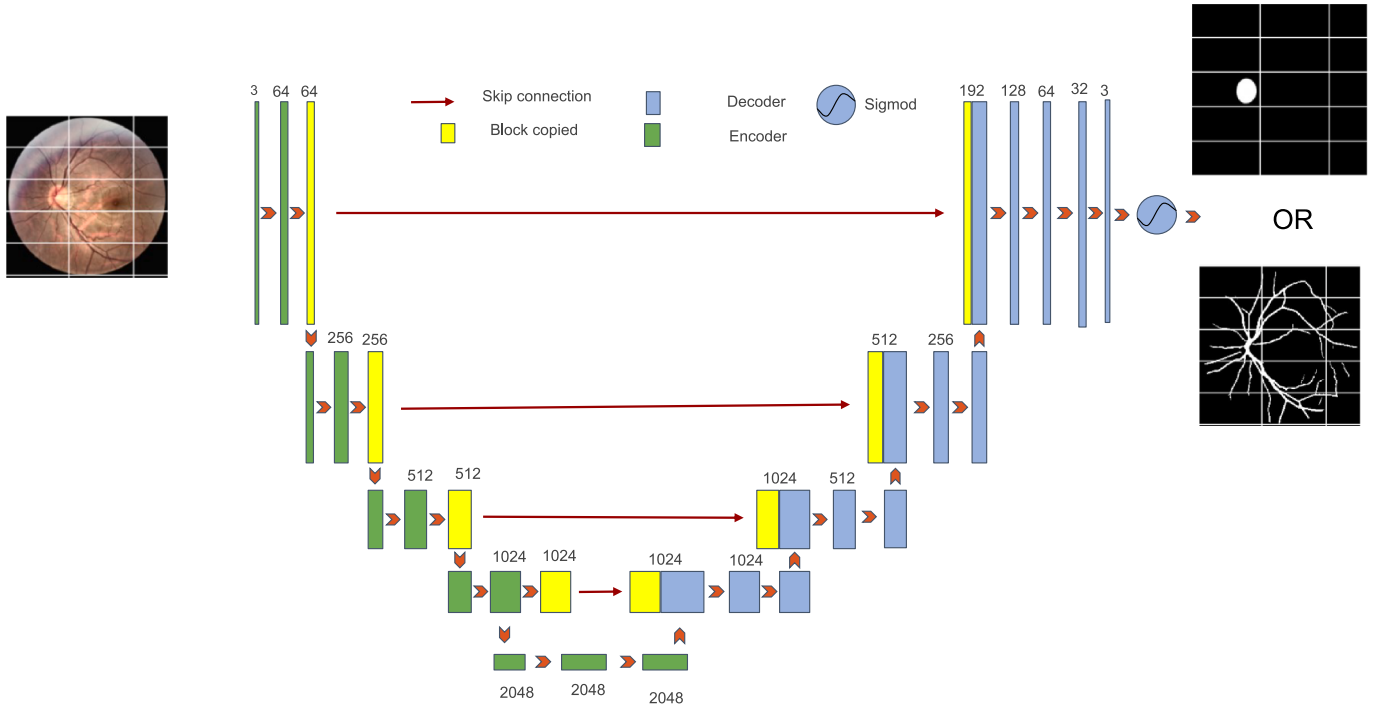


Fig. 3. Architecture of the segment module. The segment module is an encoder-decoder-based network. The encoder part consists of multiple bottleneck blocks to extract features, and the decoder part is composed of convolutional layers and Pixel Shuffle to reconstruct the extracted features into an image.

The pseudo-code of the proposed architecture is shown in Algorithm 1.

Algorithm 1 Pseudo-Code of Segment Network Process

Pre-processing:
 $image_resized = \text{resize}(image, (512, 512))$
 $image_enhanced = \text{CLAHE}(image_resized)$
 Encoding:
 $E1 = \text{MaxPooling2D}((\text{ReLU}(\text{BatchNorm}(\text{Conv2D}(image_enhanced))))$
 $E2 = \text{Conv2D}((\text{BottleneckBlock}^3(E1)))$
 $E3 = \text{Conv2D}((\text{BottleneckBlock}^4(E2)))$
 $E4 = \text{Conv2D}((\text{BottleneckBlock}^{23}(E3)))$
 $E5 = \text{BottleneckBlock}^3(E4)$

(continued on next column)

(continued)

Algorithm 1 Pseudo-Code of Segment Network Process

Decoding:
 $D5 = \text{concatenate}(\text{PixelShuffle}(\text{Conv2D}(E5)), E4)$
 $D4 = \text{concatenate}(\text{PixelShuffle}(\text{Conv2D}(D5)), E3)$
 $D3 = \text{concatenate}(\text{PixelShuffle}(\text{Conv2D}(D4)), E2)$
 $D2 = \text{concatenate}(\text{PixelShuffle}(\text{Conv2D}(D3)), E1)$
 $D1 = \text{PixelShuffle}(\text{Conv2D}(D2))$
 $OUTPUT = \text{Conv2D}(D1)$

The encoders primarily use the bottleneck blocks for construction. Fig. 4 shows the architecture of the bottleneck blocks. The bottleneck block offers advantages in reducing computational cost and improving model

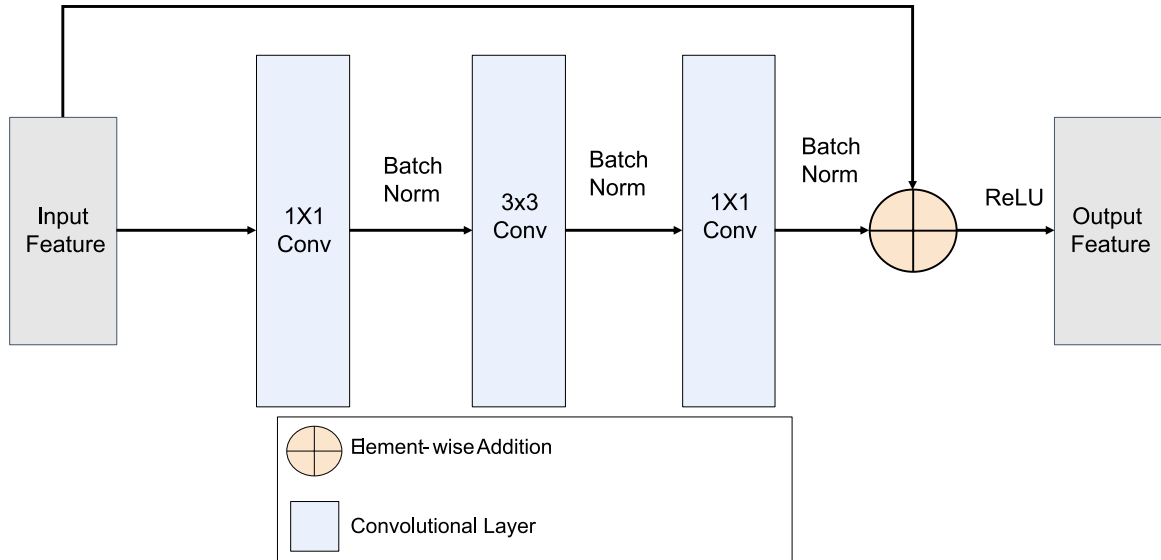


Fig. 4. Architecture of the bottleneck blocks. A bottleneck block is an efficient network architecture design. It consists of two 1×1 convolutional layers and one 3×3 convolutional layer and uses residual connections to link the input and output.

efficiency by using fewer parameters while maintaining performance. It is a widely utilized module in various deep learning architectures due to its ability to effectively capture and compress feature representations. Using convolutional kernels of different sizes, it can extract image features at various scales. Within every bottleneck block, there are two layers of 1×1 convolution, followed by a single 3×3 convolutional layer, with batch normalisation implemented subsequent to each convolution. This is followed by residual connections with the input to mitigate the vanishing gradient problem. Subsequently, the rectified linear unit (ReLU) activation function is applied as the final step.

Algorithm 2 shows the pseudo-code of the bottleneck block, where (1, 1) indicates the kernel size.

Algorithm 2 Pseudo-Code of Bottleneck block Process

```

x1 = ReLU(BatchNorm(Conv2D((1, 1))(input_tensor)))
x2 = ReLU(BatchNorm(Conv2D((3, 3))(x1)))
x3 = ReLU(BatchNorm(Conv2D((1, 1))(x1)))

```

The first encoder consists of a convolution layer with a 7×7 kernel, stride of 2 and padding of 3, along with batch normalisation and the ReLU activation function. The second encoder consists of three bottleneck blocks. The third encoder consists of 4 bottleneck blocks, the fourth encoder consists of 23 bottleneck blocks and the last encoder includes 3 bottleneck blocks. At the end of each encoder layer, the number of channels is doubled through convolution operations, and the spatial resolution is halved compared with the original.

As shown in Fig. 5 (a), in the decoder, the input feature vectors first pass through a convolutional layer that quadruples the number of channels. Then, using Pixel Shuffle, the number of channels is reduced to half of the input and the resolution is doubled. This is followed by a skip connection that connects these feature vectors with the corresponding encoder feature vectors. Finally, an up-sampling operation is performed using a convolutional layer with channel attention.

As shown in Fig. 5 (b), we use Pixel Shuffle in the decoding process as a replacement for the commonly used deconvolution operation. Pixel

Shuffle significantly improves the resolution of low-resolution images by rearranging pixels, resulting in clearer and higher-quality images. Compared with traditional up-sampling methods such as bilinear interpolation or deconvolution, the Pixel Shuffle operation is more efficient and reduces computational complexity.

In the segment task, we used the Dice loss as loss function, which is a commonly used objective function. Dice loss is a metric for evaluating model performance in binary classification tasks, particularly suited for medical image segmentation. Mathematically, the Dice loss function is defined as

$$Dice\ Loss = 1 - 2 \sum_{i=1}^N p_i g_i / \left(\sum_{i=1}^N p_i + \sum_{i=1}^N g_i \right) \quad (1)$$

In Eq. (1), p_i denotes the predicted probability of the i -th pixel, g_i denotes the ground truth label of the i -th pixel and N is the total number of pixels. Dice loss measures model performance by calculating the overlap between the predicted results and the ground truth labels. The greater the overlap is, the higher is the Dice coefficient and the lower is the Dice loss, indicating better model performance.

3.2. Glaucoma detection

In the detection task, we used the ResNet50 network architecture and modified the output of the last fully connected layer to perform glaucoma detection. We used three separate ResNet50 networks to learn different features of the fundus images. ResNet50 is widely used in image classification tasks and has been proven to be highly effective. Its deep architecture allows it to capture complex patterns and features, achieving high accuracy in image recognition and classification. It has demonstrated excellent performance in various applications. In [15], ResNet50 and methods using ResNet50 as a backbone have also shown excellent performance in glaucoma detection and OD segmentation tasks. Therefore, we also use ResNet50 as the model in our glaucoma detection module.

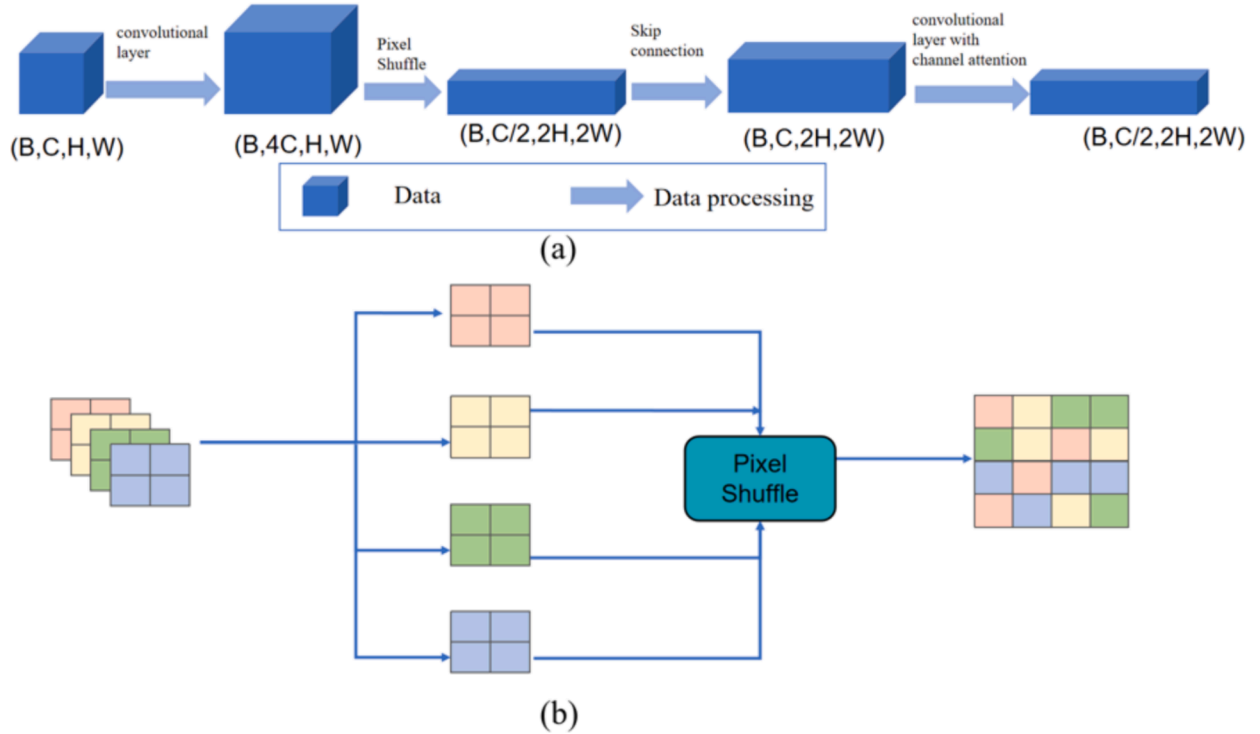


Fig. 5. Detailed structure of the decoder and the key technique, Pixel Shuffle. In particular, Fig. 5 (a) shows the detailed structure of the decoder, which consists of multiple convolutional layers and Pixel Shuffle. Fig. 5 (b) demonstrates the Pixel Shuffle operation, where pixels from four different channels are merged into a single pixel, thereby completing the decoding process.

Fig. 6 shows the architecture of ResNet50, where the Conv1 stage includes a 7×7 convolutional layer with 2 strides. The Layer1, Layer2, Layer3 and Layer4 stages consist of 3, 4, 6 and 3 bottleneck layers, respectively.

The three branches of the detection task come from the original fundus image, the segmented OD image and the segmented vessel image. To leverage the features obtained from the previous segmentation tasks, Multi-GlaucNet also applies the generated OD and vessel masks to the original fundus image. This is achieved by the element-wise multiplication of the masks with the corresponding regions of the original image.

By multiplying the mask values with the corresponding pixel values of the fundus image, the areas of interest (OD and vessels) are emphasised while suppressing the rest of the image. This step allows Multi-GlaucNet to focus on relevant regions during the subsequent detection tasks, effectively using the localised features extracted from the segmentation process. These images are then used as inputs to the respective detection branches. In addition, to ensure that features from the other regions of the fundus image are not lost and to enhance detection performance, the original image is supplied into the detection branches.

To address the class imbalance issue in the detection task of Multi-GlaucNet, we use the focal loss (2) method, which tackles this problem by introducing a balance coefficient (α) and an adjusting parameter (γ) as shown in Eq. (2). In Eq. (2), C denotes the total number of categories and t_i denotes the element corresponding to the i -th category of the true label. Typically, only one element is set to 1 (indicating the true category), and the rest are set to 0.

$$\text{Focal Loss}(y, t) = -\alpha \sum_{i=1}^C t_i (1 - y_i)^\gamma \log(y_i) \quad (2)$$

In Eq. (2), y_i denotes the probability assigned by the model to the i -th category in its prediction and γ is a regularisation parameter that governs the weighting of samples based on their level of difficulty. A higher value of γ increases the weight assigned to challenging-to-classify samples, thereby mitigating the influence of easily classified samples. In the detection module, we set the balance α to 3 and the adjusting parameter γ to 10.

4. Experimental results

4.1. Datasets

In our experiments, we performed training for the OD segmentation task on the REFUGE [18], ACRIMA [19], ORIGA [20] and DRISHTI-GS [21] datasets and compared the test results with other deep learning models for segmentation tasks. For vessel segmentation, we created a dataset by combining data from the FIVES [22], DRIVE [23], CHASE_DB1 [24], STARE [25] and HRF [26] datasets. Table 1 presents the different datasets used in the experiments.

For the OD segmentation task, we trained on the ORIGA, DRISHTI-



Fig. 6. ResNet50 architecture.

Table 1

Overview of used datasets.

Dataset Name	Number of Images	Categories	Purpose
REFUGE train/validation set	800	With/Without Glaucoma	OD Segmentation Glaucoma Detection
ACRIMA	705	With/Without Glaucoma	Glaucoma Detection
ORIGA	650	With/Without Glaucoma	OD Segmentation
DRISHTI-GS	101	With/Without Glaucoma	OD Segmentation
FIVES	800	Not Specified	Vessels Segmentation
DRIVE	40	Not Specified	Vessels Segmentation
CHASE_DB1	56	Not Specified	Vessels Segmentation
STARE	20	Not Specified	Vessels Segmentation
HRF	45	Not Specified	Vessels Segmentation

GS and REFUGE datasets. In the vessel segmentation task in fundus images, to ensure that the model performs well across different datasets, we chose to merge multiple fundus image datasets used for vessel segmentation. We then trained the model on the merged dataset. In addition, to compare our proposed method with other vessel segmentation approaches and demonstrate its performance, we conducted training on individual datasets. For the detection task, we conducted training on the REFUGE and ACRIMA datasets. Table 2 presents the data distribution during the training, validation and testing processes.

4.2. Performance evaluation

In our experiments, we used the Dice coefficient to evaluate the OD segmentation task. To evaluate the blood vessel segmentation and glaucoma detection tasks, we adopted five metrics in different task results: accuracy, F1 score, sensitivity, specificity and area under the curve (AUC).

In these evaluations, TP denotes true positives; TN, true negatives; FP, false positives; and FN, false negatives.

The Dice coefficient is a commonly used evaluation metric to measure the similarity or overlap between two sets. In image segmentation tasks, the Dice coefficient is often used to assess the consistency between the segmentation result and the ground truth labels. Eq. (3) shows the formula for the Dice coefficient.

$$\text{Dice} = 2 \times TP / (2 \times TP + FP + FN) \quad (3)$$

Accuracy measures the overall detection correctness of the model on the entire sample set. Eq. (4) presents the formula for accuracy.

Table 2

Data distribution. In glaucoma detection, G denotes the number of fundus images with glaucoma and N denotes the number of normal fundus images.

Dataset	Train set 70 %	Test set 15 %	Validation set 15 %
OD Segmentation			
ORIGA	488	81	81
DRISHTI-GS	71	15	15
REFUGE	560	120	120
Vessel Segmentation			
CHASE_DB1	20	4	4
STARE	14	3	3
Mix Dataset	671	145	145
Glaucoma Detection			
REFUGE	G:57 N:503	G:14 N:106	G:9N:111
ACRIMA	G:330 N:257	G:33 N:26	G:33 N:26

$$\text{Accuracy} = (TP + TN)/(TP + TN + FP + FN) \quad (4)$$

The F1 score is a comprehensive evaluation metric that considers both precision and recall of a model. Eq. (5) presents the formula for F1 score.

$$\text{F1 Score} = 2 \times (\text{precision} \times \text{recall})/(\text{precision} + \text{recall}) \quad (5)$$

Sensitivity represents the proportion of actual positive samples that are correctly predicted as positive. Eq. (6) presents the formula for sensitivity.

$$\text{Sensitivity} = TP/(TP + FN) \quad (6)$$

Specificity represents the proportion of actual negative samples that are correctly predicted as negative. Eq. (7) presents the formula for specificity.

$$\text{Specificity} = TN/(TN + FP) \quad (7)$$

The AUC refers to the area under the receiver operating characteristic curve of a detection model, which is used to measure the model's ability.

4.3. Experimental environment

Our experiments were conducted on a computer equipped with a 12th Gen Intel(R) Core (TM) i5-12400F processor and an NVIDIA GeForce GTX 1660 SUPER with 6 GB of VRAM.

We used Python version 3.11.3 and PyTorch version 2.0.1. In our experiments, we used a batch size of 1. During testing, we used a batch size of 1 with an epoch of 100. The learning rate used in the training was $1.909e - 04$. During the training process of the aforementioned tasks, we used the Adam optimisation algorithm during training and the exponential decay learning rate scheduling method to adjust the learning rate in the optimiser. After each training iteration, the learning

rate was adjusted to 0.98 times its original value.

4.4. Results

In this section, we present a comparison of various individual tasks with recent methods across different datasets.

For the OD segmentation task, we compared our results on the ORIGA, REFUGE and DRISHTI-GS datasets. For the vessel segmentation task, comparisons were made using the CHASE_DB1 and STARE datasets. We also show the result on the mix dataset. Fig. 7 shows the learning curves for the OD segmentation task and vessel segmentation task across different datasets. From Fig. 7(A) and 7(B), it can be seen that during the training process of the OD segmentation task, the accuracies and losses for both the training and validation sets tend to stabilize after 20 epochs on both datasets, with the losses steadily decreasing. Fig. 7(C) reveals that when training on a single vascular segmentation dataset, the training set accuracy continuously and steadily increases, while the loss steadily decreases. The Dice coefficient and loss of the validation set, however, tend to stabilize after 50 epochs. Fig. 7(D) indicates that when training on a mixed dataset, the training Dice coefficient continuously rises, but the validation set Dice coefficient fluctuates, suggesting that the training outcomes are not optimal under conditions of large data volume and cross-domain scenarios, indicating the need for more iterations of training.

For the glaucoma detection task, we evaluated our approach using the REFUGE and ACRIMA datasets. Fig. 8 shows the learning curves for the glaucoma detection task across different datasets. From Fig. 8(A), it can be observed that on the ACRIMA dataset, both training and validation accuracies essentially stabilize after 20 epochs, while the training loss experiences a significant increase around 60 epochs, followed by a subsequent decline. In contrast, Fig. 8(B) demonstrates that on the REFUGE dataset, training and validation accuracies stabilize after approximately 30 epochs, and both training and validation losses tend to

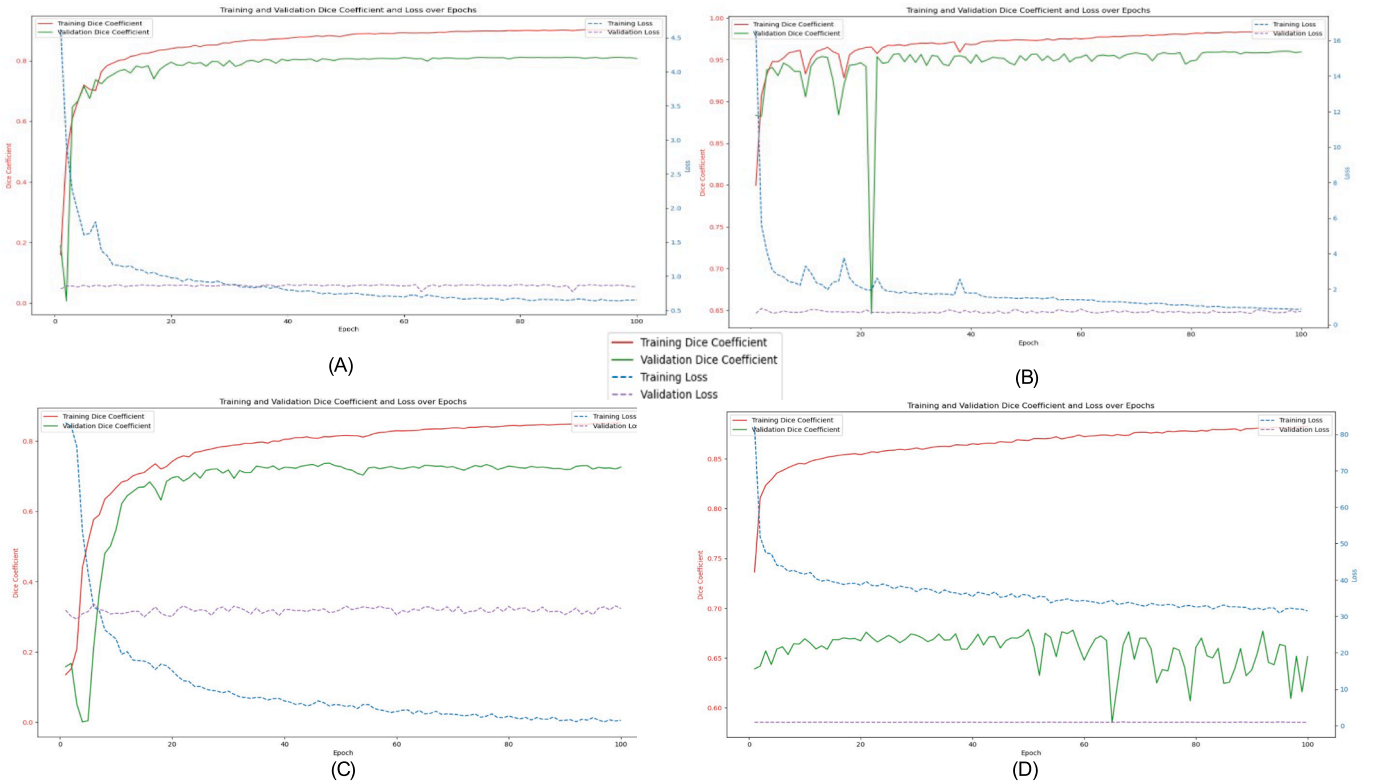


Fig. 7. Learning curves of Dice scores/losses for the training and validation sets of the segmentation task across different datasets, with the vertical axis representing the Dice score and the horizontal axis representing the epochs. (A) and (B) depict the learning curves for the OD segmentation task on the REFUGE and ORIGA datasets, respectively. (C) and (D) show the learning curves for the vessel segmentation task on the CHASE_DB1 and mixed datasets, respectively.

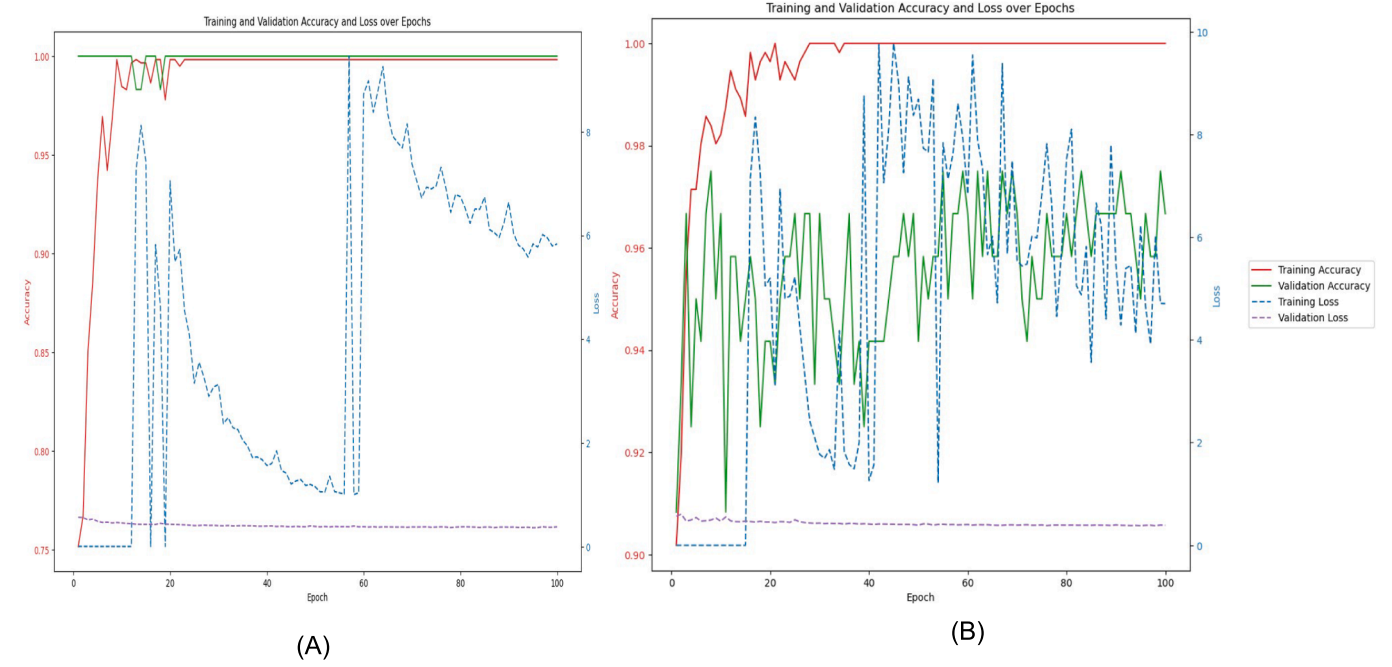


Fig. 8. Learning curves of accuracy/losses for the training and validation sets of the glaucoma detection task across different datasets, with the vertical axis representing the Dice score and the horizontal axis representing the epochs. (A) and (B) depict the learning curves on the ACRIMA and REFUGE datasets.

fluctuate and converge during the training process.

Table 3 presents the results of our proposed Multi-GlaucNet compared with other recent OD segmentation methods. As shown in Table 3, our proposed Multi-GlaucNet achieved Dice scores of 96.7 %, 95.9 % and 96.6 % on the ORIGA, REFUGE and DRISHTI-GS datasets, respectively. Compared with other recent single-task models for OD segmentation, our proposed method performs similarly to the best results and demonstrates stable and good performance across different datasets.

In the vessel segmentation task, we used a mixed dataset. To compare with other blood vessel segmentation networks, we trained and tested on the CHASE_DB1 and STARE dataset. Table 4 presents the training results and a comparison with existing vessel segmentation models.

In Table 4, we list the sensitivity, specificity, accuracy, AUC and F1 score of different methods for comparison. On the CHASE_DB1 and STARE datasets, our proposed Multi-GlaucNet performs on par with the latest fundus vessel segmentation methods across various metrics and

achieves high segmentation results on the merged dataset. This demonstrates that our proposed method can accurately segment blood vessels in fundus images.

We compared our results with other dedicated detection networks, as shown in Table 5. As shown in Table 5, on the REFUGE and ACRIMA datasets, our proposed Multi-GlaucNet achieved accuracies of 0.967 and 0.985, and AUC of 0.950 and 0.981, respectively. Compared with the recent glaucoma detection methods listed in the table, our method consistently performed better.

Fig. 9 presents the confusion matrix and AUC curves for the glaucoma detection task on two different datasets. Fig. 9(A) and Fig. 9(B) shows the confusion matrix and AUC curves on REFUGE dataset. Fig. 9 (C) and Fig. 9(D) shows the confusion matrix and AUC curves on ACRIMA dataset. From Fig. 9(A) and (C), it can be observed that the model exhibits a high accuracy in detecting positive cases of glaucoma, but there are instances where negative cases are incorrectly identified as glaucoma. Fig. 9(B) and (D) demonstrate that the model has a high AUC score, indicating its ability to accurately detect glaucoma.

We compared the proposed Multi-GlaucNet with several other multi-task deep learning models tailored for glaucoma, and the corresponding results are shown in Table 6. As shown in Table 6, Multi-GlaucNet achieved a high Dice score for the OD segmentation task and similar superior performance for glaucoma detection. In addition, the proposed Multi-GlaucNet possesses a blood vessel segmentation ability, which is not found in other models. Thus, the proposed technique enhances glaucoma detection performance based on its capability in blood vessel segmentation tasks.

To ensure that each module in Multi-GlaucNet contributes effectively to glaucoma detection, we conducted ablation experiments. In the ablation experiments, we focused on three different modules in Multi-GlaucNet. We referred to the module with the original image input as ORI, the module with the input of the OD segmentation image as OD and the module with the input of the vessel segmentation image as VESSEL. Table 7 presents the test results of each module, the combination of OD and VESSEL and the combination of all three modules. Given the presence of two distinct sample imbalances in the REFUGE dataset, our ablation studies focused more on the enhancement of the AUC metric. As illustrated in Table 7, the culmination of this approach in a three-branch

Table 3

OD segmentation results. The best-performing structures within each dataset are highlighted in bold.

Methods	Dice Disc (%)	Dataset
U-Net	95.0	ORIGA
M-Net [27]	96.8	
GRBNet [27]	97.7	
MT [28]	94.3	
UAMT [29]	95.2	
URPC [30]	96.0	
CABNet [31]	96.2	
Shyamalee [32]	93.0	
Multi-GlaucNet (Ours)	96.7	
U-Net	93.1	
M-Net [33]	93.6	REFUGE
PKRT-Net [3]	97.5	
ET-Net [34]	95.3	
Shyamalee [32]	96.4	
Multi-GlaucNet (Ours)	95.9	
PKRT-Net [3]	97.6	DRISHTI-GS
Shyamalee [32]	86.4	
Multi-GlaucNet (Ours)	96.6	

Table 4

Blood vessel segmentation results.

Methods	Sensitivity	Specificity	Accuracy	AUC	F1 Score	Dataset
AGNet [35]			0.9743	0.9863	—	CHASE_DB1
LadderNet [36]	0.7978	0.9818	0.9656	0.9839	0.8031	
AACA-MLA-D-UNet [37]	0.8402	0.9801	0.9673	0.9874	0.8248	
Wave-Net [38]	0.8284	0.9821	0.9964	—	0.8349	
TUnet-LBF [39]	0.8004	0.9806	0.9708	—	0.7896	
SegR-Net [40]	0.8329	0.9838	—	—	0.8030	
ARSA-Unet [41]	0.8429	0.9753	0.9631	—	0.8074	STARE
Multi-GlaucNet(Ours)	0.9304	0.9987	0.9964	0.9832	0.8102	
AACA-MLA-D-UNet [37]	0.7914	0.9870	0.9665	0.9864	0.8276	
Wave-Net [38]	0.9641	0.9836	0.7902	—	0.8140	
TUnet-LBF [39]	0.8004	0.9852	0.9681	—	0.8198	
SegR-Net [40]	0.8212	0.9741	—	—	0.8075	
ARSA-Unet [41]	0.8793	0.9772	0.9663	—	0.8535	Mix Dataset
Multi-GlaucNet (Ours)	0.7842	0.9981	0.9880	0.9850	0.7854	
Multi-GlaucNet (Ours)	0.8936	0.9943	0.9798	0.9874	0.8562	

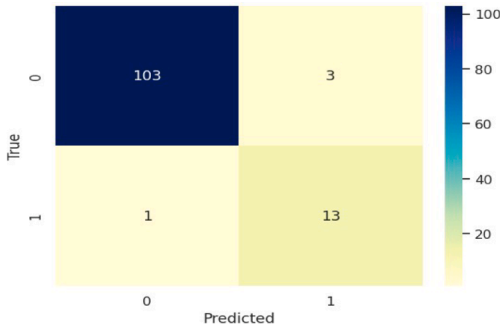
Table 5

Glaucoma detection results.

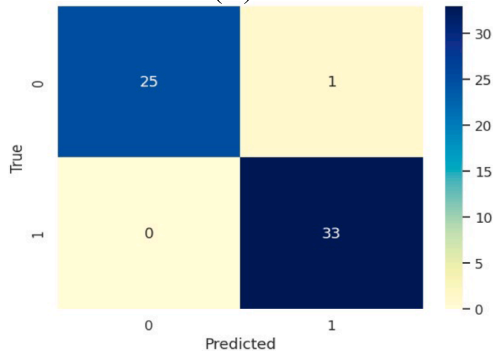
Methods	Sensitivity	Specificity	Accuracy	AUC	Dataset
Sreng [42]	—	—	0.957	0.951	REFUGE
ResNet50 [17]	—	—	0.935	0.930	
Shyamalee [33]	0.937	0.846	0.911	0.930	
Multi-GlaucNet (Ours)	0.928	0.971	0.967	0.950	
Shyamalee [32]	0.978	0.933	0.972	0.980	ACRIMA
Sonti [43]	0.966	0.966	0.967	0.967	
ResNet50 [17]	0.982	0.833	0.956	0.960	
Multi-GlaucNet (Ours)	1.000	0.961	0.985	0.981	

amalgamation yields an AUC score of 0.9547, the highest recorded, demonstrating that each branch contributes to enhancing the precision of glaucoma detection.

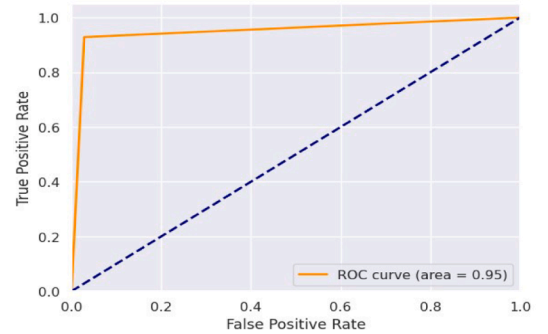
As shown in Table 8, we also conducted ablation studies on different loss functions and up-sampling methods during the training process. The results indicate that among the three commonly used loss functions, training with Dice loss yielded the best outcome, achieving 96.9 %. In the ablation study on up-sampling operations, the Pixel Shuffle operation showed a 0.5 % improvement compared with the traditional transposed convolution operation, demonstrating the superiority of our chosen method. In the detection loss function ablation study, we compared the focal loss using the commonly used cross entropy loss, demonstrating the efficiency of our chosen loss function. In the channel attention ablation study, we also showed how the channel attention mechanism in the decoder enhances the performance of the segmentation task.



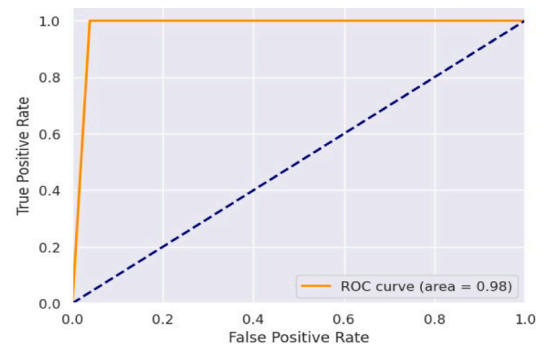
(A)



(C)



(B)



(D)

Fig. 9. Confusion matrix and AUC curve on the REFUGE and ACRIMA datasets. Fig. 9 (A) shows the confusion matrix results on REFUGE, and Fig. 9 (B) shows the AUC curve results on REFUGE. Fig. 9 (C) shows the confusion matrix results on ACRIMA, and Fig. 9 (D) shows the AUC curve results on ACRIMA.

Table 6

Comparison with multi-task models on the REFUGE dataset.

Methods	OD Dice	Glaucoma Detection Accuracy	Glaucoma Detection AUC	Blood Vessel Segment	Fovea Detection
Hervella et al. [14]	0.959	–	0.976	×	×
Pascal et al. [9]	0.951	–	0.961	×	✓
Wang et al. [44]	0.960	–	0.964	×	×
Multi-GlaucNet (ours)	0.959	0.967	0.950	✓	×

Table 7

Ablation experiments on REFUGE. In the ablation study, we randomly selected 720 images as the training set and 80 images as the validation set.

Methods	Accuracy	AUC	F1 Score
ORI	0.963	0.9012	0.8235
OD	0.969	0.9129	0.8148
VESSEL	0.944	0.8975	0.5263
OD+VESSEL	0.963	0.9306	0.8750
ORI+VESSEL+ORI	0.988	0.9547	0.9091

Table 8

Ablation study on different loss functions, various up-sampling operations and channel attention in the decoding process on the DRISHTI-GS dataset. The detection loss function ablation study was conducted on the ACRIMA dataset.

Segment Loss Function Ablation Study	
Loss Function	Dice Score (%)
Dice Loss	96.6
MSE Loss	95.5
IoU Loss	96.5
Detection Loss Function Ablation Study	
Loss Function	Accuracy
Focal Loss	0.985
Cross Entropy Loss	0.949
Up-sampling Operation Ablation Study	
Up-sampling Operation	Dice Score (%)
Pixel Shuffle	96.6
Transposed Convolution	96.1
Channel Attention Ablation Study	
With/without Channel Attention	Dice Score (%)
With Channel Attention	96.6
Without Channel Attention	96.3

5. Conclusion

In this paper, we proposed a multi-task deep learning model called Multi-GlaucNet that can simultaneously perform OD segmentation, vessel segmentation and glaucoma detection on fundus images. This model not only enables automatic glaucoma detection in fundus images but also provides segmentation of blood vessels and OD to assist in diagnosis, improving the accuracy of glaucoma detection from multiple aspects.

In the Multi-GlaucNet framework, OD segmentation and retinal blood vessel segmentation modules are architecturally designed with an array of bottleneck blocks functioning as an encoder. This encoder is responsible for extracting salient features from fundus images. Subsequently, the decoding process incorporates Pixel Shuffle, substituting traditional transposed convolution techniques, supplemented by a channel-wise attention mechanism. This mechanism is instrumental in transforming high-dimensional feature representations into refined final mask images. Furthermore, the glaucoma detection module of Multi-

GlaucNet leverages the capabilities of ResNet50 for enhanced feature extraction, culminating in the execution of the classification task.

For the OD segmentation task, we achieved the highest Dice score of 96.7 % on the ORIGA dataset. For the vessel segmentation task, we achieved an accuracy of 0.9798, an AUC of 0.9874 and an F1 score of 0.8562 on the mixed dataset. For the glaucoma detection task, we achieved an accuracy of 0.9280 and an AUC of 0.9500 on the REFUGE dataset. The above results demonstrate the excellent performance of the proposed Multi-GlaucNet for all three tasks and its consistent performance across different datasets. This confirms that Multi-GlaucNet exhibits outstanding performance in OD segmentation, vessel segmentation and glaucoma detection tasks. The proposed model demonstrates great potential for multi-task deep learning-based glaucoma diagnosis.

In clinical practice, our proposed Multi-GlaucNet supports medical professionals by offering dual diagnostic aids: automated glaucoma detection results and detailed segmentations of blood vessels and OD regions. This dual approach allows doctors to use both the model-generated outcomes and their own observations of the segmented areas for a more comprehensive diagnosis.

However, this method still has some limitations. For example, it does not fully extract other information from fundus images (such as fovea detection) to complete more tasks and improve the effectiveness of glaucoma detection. In addition, it does not use other modalities (such as OCT images) to enhance the final glaucoma detection results. These are the shortcomings of this method and the directions for future improvements. In future work, we will consider a more comprehensive approach to CAD of glaucoma from the perspectives of additional tasks and different modalities.

CRedit authorship contribution statement

Haoren Xiong: Writing – original draft, Methodology, Investigation, Formal analysis, Conceptualization. **Fei Long:** Validation, Investigation, Formal analysis. **Mohammad S. Alam:** Writing – review & editing, Conceptualization. **Jun Sang:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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